

KARMA RIDE:

Fair Allocation of City-Center Access to Ride-Hailing Drivers Without Money

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Taxi and Ride-Hailing Drivers Gravitare Toward the City Center



Traffic Congestion



Underserved Periphery

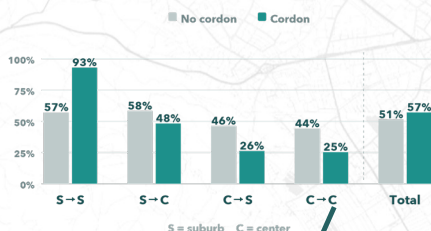
Setting Cordon Areas

Limit taxi access to the center: during rush hours, taxis can enter only if they are

permitted ●

Total number of permits is fixed as **N**. They are a scarce resource to be allocated.

Service Rate Comparison



Demand in the city center is encouraged to shift to public transportation (green & efficient)



Check out our website for more technical details and next steps



Who Gets a Permit?

Money? Favors drivers with greater ability to pay.

Lottery? Only fair ex-ante. Repeated bad luck is possible.

Rotation? Fair but ignores who needs the permit most today.

Artificial Currency: Karma

Bidding

Allocation

Sept. 4rd, Fri
Balance: 18 KP
Bid: KP
Bid: KP
Bid: KP

Highest N bidders win permits (e.g., N = 2)

Redistribution

Redistribute the collected KPs (e.g., $9 \times 2/3 = 6$ KP each)

Payment

Winners pay the lowest bid of the winners (e.g., 9 KP)

Drop-out Rate in 30 Days

(proportion of drivers leaving the market if average income is lower than a threshold in X days)

Patience window	Karma	Lottery
7 days	5.1%	10.3%
14 days	1.5%	4.6%
21 days	0.7%	2.6%
28 days	0.3%	1.2%

Average Income Dispersion (Inequality)



Karma provides more stable income, better equality and is not affected by external wealth